

1. Objective

This practical demonstrates Flutter screen navigation using `Navigator.push`, `Navigator.pop`, and `MaterialPageRoute`. It also shows how to pass a product name from a product list to a `ProductDetailScreen`, similar to a simple Flipkart-style flow.

2. Screen Flow

MainScreen → CartScreen → OffersScreen

MainScreen → ProductDetailScreen

The device/system back button automatically pops the current route, so pressing it on OffersScreen returns the user to CartScreen.

3. Important Navigation Concepts

Concept	Purpose
<code>Navigator.push</code>	Adds a new screen/route on top of the navigation stack.
<code>Navigator.pop</code>	Removes the current screen and returns to the previous screen.
<code>MaterialPageRoute</code>	Creates a Material Design route for screen navigation.
Constructor argument	Passes data, such as a product name, into another screen.

```

import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedStackBanner: false,
      title: 'Navigation Demo',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
    );
  }
}

// 1. MainScreen - List of products to be in Cart
class MainScreen extends StatelessWidget {
  final List<String> products = [
    'Puma Nrgo 70 Pro',
    'boAt Rockerz S10',
    'Apple iPhone 13'
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Puma Nrgo 70 Pro')),
      body: ListView.builder(
        itemCount: products.length,
        itemBuilder: (context, index) {
          return ListTile(
            title: Text(products[index]),
            style: ListTileStyle.horizontal,
            trailing: Icon(Icons.arrow_forward_ios),
            onTap: () {
              Navigator.push(
                context,
                MaterialPageRoute(builder: (context) => ProductDetailScreen(
                  products[index],
                )),
              );
            },
          );
        },
      ),
    );
  }
}

// 2. ProductDetailScreen - Product details
class ProductDetailScreen extends StatelessWidget {
  final String productName;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Product Details')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Image.asset(
              'assets/images/boAtRockerzS10.png',
              width: 100,
              height: 100,
            ),
            Text(
              'boAt Rockerz S10',
              style: TextStyle(fontWeight: FontWeight.bold,
                fontSize: 24,
              ),
            ),
            Text(
              'This is the detail page of boAt Rockerz S10',
              style: TextStyle(fontWeight: normal,
                fontSize: 16,
              ),
            ),
          ],
        ),
      ),
    );
  }
}

// 3. CartScreen - Your Cart
class CartScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Your Cart')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text(
              'Your cart is empty!',
              style: TextStyle(fontWeight: bold,
                fontSize: 18,
              ),
            ),
            TextButton(
              onPressed: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(builder: (context) => MainScreen()),
                );
              },
              child: Text('Go to Main'),
            ),
          ],
        ),
      ),
    );
  }
}

// 4. OffersScreen - Special Offers
class OffersScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Offers')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Image.asset(
              'assets/images/specialOffer.png',
              width: 100,
              height: 100,
            ),
            Text(
              'Special Offers for You!',
              style: TextStyle(fontWeight: bold,
                fontSize: 18,
              ),
            ),
            Text(
              'Up to 50% Off on Selected Items',
              style: TextStyle(fontWeight: normal,
                fontSize: 16,
              ),
            ),
          ],
        ),
      ),
    );
  }
}

```

